

**Name : Fahad Ali**

**Roll no : 23P-0569**

**Section : BCS-4k**

**Lab 1 Task**

```

student@Lab5-Pc:~/Desktop$ cd 23P-0569
student@Lab5-Pc:~/Desktop/23P-0569$ ls
student@Lab5-Pc:~/Desktop/23P-0569$ arch
x86_64
student@Lab5-Pc:~/Desktop/23P-0569$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:           39 bits physical, 48 bits virtual
Byte Order:              Little Endian
CPU(s):                  16
On-line CPU(s) list:     0-15
Vendor ID:               GenuineIntel
Model name:              11th Gen Intel(R) Core(TM) i7-11700 @ 2.50GHz
CPU family:              6
Model:                   167
Thread(s) per core:      2
Core(s) per socket:      8
Socket(s):               1
Stepping:                1
CPU max MHz:             4900.0000
CPU min MHz:             800.0000
BogoMIPS:                4992.00
Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall nx pdpe1gb rdtscp ln c
onstant_tsc art arch_perfmon pebs bts rep_good nopl xtopology nonstop_tsc cpuid aperfperf tsc_known_freq pni pclmulqdq dtes64 monitor ds_cpl vnx smx
est tm2 ssse3 sdbg fma cx16 xtpr pdcm pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand lahf_lm abm 3dnowprefetch cp
uid_fault epb ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow flexpriority ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid npx avx5
12f avx512dq rdseed adx snap avx512ifma clflushopt intel_pt avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves dtherm ida arat pln pts h
wp hwp_notify hwp_act_window hwp_epp hwp_pkg_req vnni avx512vbni unip pku ospke avx512_vbni2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpo
pcntdq rdpid fsrm md_clear flush_l1d arch_capabilities

Virtualization features:
Virtualization:          VT-x
Caches (sum of all):
L1d:                     384 KiB (8 instances)
L1i:                     256 KiB (8 instances)
L2:                      4 MiB (8 instances)
L3:                      16 MiB (1 instance)
NUMA:
NUMA node(s):            1
NUMA node0 CPU(s):       0-15
Vulnerabilities:
Gather data sampling:    Mitigation; Microcode
Itlb multihit:          Not affected
L1tf:                   Not affected
Mds:                    Not affected
Meltdown:               Not affected
Mmio stale data:         Mitigation; Clear CPU buffers; SMT vulnerable
Reg file data sampling:  Not affected
Retbleed:                Mitigation; Enhanced IBRS
Spec rstack overflow:    Not affected
Spec store bypass:       Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:              Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:              Mitigation; Enhanced / Automatic IBRS; IBPB conditional; RSB filling; PBSRB-eIBRS SW sequence; BHI SW loop, KVM SW loop
Srbds:                  Not affected
Tsx async abort:         Not affected

```

```

student@Lab5-Pc:~/Desktop/23P-0569$ uname -r
Command 'uname' not found, did you mean:
  command 'uuname' from deb uucp (1.07-27build3)
  command 'uniname' from deb uniutils (2.27-2build1)
  command 'uname' from deb coreutils (8.32-4.1ubuntu1.2)
Try: sudo apt install <deb name>
student@Lab5-Pc:~/Desktop/23P-0569$ uname -r
6.8.0-65-generic
student@Lab5-Pc:~/Desktop/23P-0569$ date
Fri Jan 23 10:13:18 AM PKT 2026
student@Lab5-Pc:~/Desktop/23P-0569$ sudo date 01231030
Fri Jan 23 10:30:00 AM PKT 2026
student@Lab5-Pc:~/Desktop/23P-0569$

```

```
grep[0-9]: command not found
student@Lab5-Pc:~$ ls | grep [0-9]
1.png
23k-2063final
23k-2063final.zip
2.png
file1.txt
oslab2
OSSPRING2025
task1.txt
student@Lab5-Pc:~$
```

```
student@Lab5-Pc:~/Desktop$ cd 23P-0569
student@Lab5-Pc:~/Desktop/23P-0569$ mkdir dir1
student@Lab5-Pc:~/Desktop/23P-0569$ ls
dir1
student@Lab5-Pc:~/Desktop/23P-0569$ mkdir dir2 dir3
student@Lab5-Pc:~/Desktop/23P-0569$ ls
dir1 dir2 dir3
student@Lab5-Pc:~/Desktop/23P-0569$ pwd
/home/student/Desktop/23P-0569
student@Lab5-Pc:~/Desktop/23P-0569$ rm file1
```

```
student@Lab5-Pc:~/Desktop/23P-0569$ rm file1
rm: cannot remove 'file1': No such file or directory
student@Lab5-Pc:~/Desktop/23P-0569$ touch file1
student@Lab5-Pc:~/Desktop/23P-0569$ rm file1
student@Lab5-Pc:~/Desktop/23P-0569$ rm -r dir1
student@Lab5-Pc:~/Desktop/23P-0569$ rmdir dir2
student@Lab5-Pc:~/Desktop/23P-0569$ touch -c -t 202601231007
```

```
touch: invalid date format '202601231007'
student@Lab5-Pc:~/Desktop/23P-0569$ touch -c -t 202601231007 dir3
student@Lab5-Pc:~/Desktop/23P-0569$
```